

Improving Maternal Health Risk Classification Using Machine Learning

Baseline Group Presentation
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Research Question & Motivation

Research Question:

- How can machine learning methods improve the accuracy and efficiency of maternal health risk classification during pregnancy?

Why It's Interesting:

- ~810 maternal and 6,700 newborn deaths daily (WHO).
- Many maternal deaths are preventable.
- Early, accurate risk prediction can save lives, especially in under-resourced regions.
- Deep learning models can run quickly on local devices, no GPU needed.



Dataset Overview

- **Source:** UCI Machine Learning Repository
- **Origin:** IoT monitoring systems in rural Bangladesh

Size:

- 1017 Samples, 7 features

Variables:

- Features: Age, SystolicBP, DiastolicBP, Blood Sugar, Body Temp, Heart Rate
- Target: RiskLevel (Low, Mid, High)

Missing Data:

- Lacks previous pregnancies, complications, miscarriages

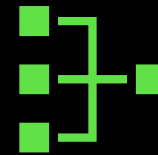
Algorithms



Baseline Model: Logistic Regression



Random Forest: Ensemble, handles feature importance well



Artificial Neural Network: Models nonlinear patterns, lightweight for local use



SVC: Support Vector Classification

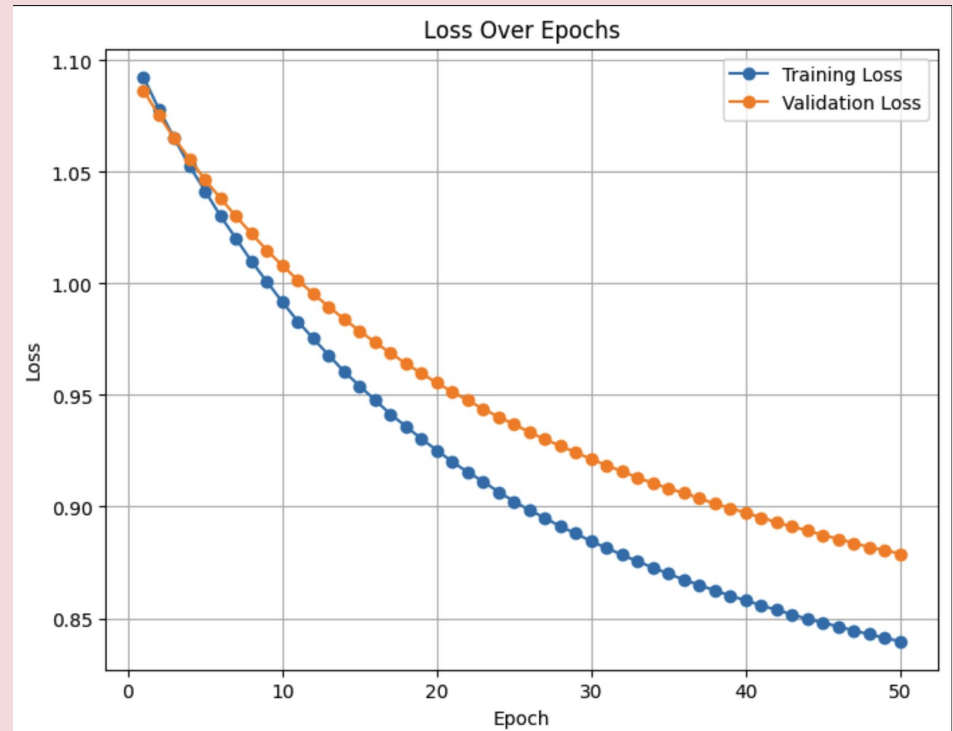
Logistic Regression - Baseline Model

Initial Parameters:

- Epochs: 50
- Batch size: 32
- Activation: softmax

Results:

- Training Accuracy: 0.6494
- Validation accuracy: 0.5889
- Training loss: 0.8396
- Validation loss: 0.8788



Logistic Regression - Optimized Model

Hyperparameter Tuning:

- Epochs tested: [10, 20, 50, 100]
- Batch sizes tested: [16, 32, 64, 128]
- Best parameters Found:
 - Epochs: 100
 - Batch Size: 32

Results:

- Training accuracy: 0.6458
- Validation accuracy: 0.6111
- Test accuracy: 0.7253

	Epochs	Batch Size	Train Accuracy	Val Accuracy	Train Loss	Val Loss
0	10	16	0.6347	0.5889	0.8191	0.8543
1	10	32	0.6421	0.5889	0.8111	0.8447
2	10	64	0.6384	0.6000	0.8074	0.8396
3	10	128	0.6384	0.6000	0.8055	0.8375
4	20	16	0.6421	0.6000	0.7940	0.8165
5	20	32	0.6458	0.6000	0.7899	0.8102
6	20	64	0.6421	0.6000	0.7883	0.8071
7	20	128	0.6458	0.6000	0.7874	0.8044
8	50	16	0.6458	0.6000	0.7817	0.7894
9	50	32	0.6458	0.6000	0.7803	0.7858
10	50	64	0.6458	0.6111	0.7798	0.7837
11	50	128	0.6458	0.6000	0.7797	0.7838
12	100	16	0.6458	0.6111	0.7791	0.7797
13	100	32	0.6458	0.6111	0.7788	0.7780
14	100	64	0.6458	0.6111	0.7787	0.7791
15	100	128	0.6458	0.5889	0.7788	0.7779

Artificial Neural Networks - Baseline Model

Initial Parameters:

- Epochs: 100
- Batch size: 32
- Hidden Layers:
 - Layer 1: 64 neurons (ReLU activation) with Dropout (30%)
 - Layer 2: 32 neurons (ReLU activation) with Dropout (20%)
- Optimizer: Adam (Learning Rate: 0.001)

Results:

- Training Accuracy: 0.8373
- Validation accuracy: 0.6406
- Test Accuracy: 0.6765

Layer (type)	Output Shape	Param #
dense_1 (Dense)	(None, 64)	448
dropout (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 32)	2,080
dropout_1 (Dropout)	(None, 32)	0
dense_3 (Dense)	(None, 3)	99

Artificial Neural Networks - Optimized Model

Hyperparameter Tuning:

- Epochs tested: [10, 20, 50, 100]
- Batch sizes tested: [16, 32, 64, 128]
- Best parameters Found:
 - Epochs: 5
 - Batch Size: 64

Results:

- Training accuracy: 0.7579
- Validation accuracy: 0.7500
- Test accuracy: 0.6471

	Epochs	Batch Size	Train Accuracy	Val Accuracy	Train Loss	Val Loss
0	5	8	0.7421	0.7344	0.6162	0.6671
1	5	16	0.7341	0.7344	0.6151	0.6664
2	5	32	0.7381	0.7500	0.6021	0.6656
3	5	64	0.7579	0.7500	0.5850	0.6651
4	10	8	0.7183	0.7344	0.6142	0.6519
5	10	16	0.7619	0.7500	0.5699	0.6715
6	10	32	0.7659	0.7188	0.5711	0.6765
7	10	64	0.7738	0.7188	0.5410	0.6737
8	20	8	0.7500	0.7031	0.5558	0.6859
9	20	16	0.7579	0.7031	0.5435	0.7158
10	20	32	0.7659	0.6562	0.5184	0.7155
11	20	64	0.7817	0.6719	0.5020	0.7165
12	50	8	0.7778	0.7031	0.4865	0.7304
13	50	16	0.8214	0.6719	0.4556	0.7671
14	50	32	0.8095	0.6562	0.4580	0.8091
15	50	64	0.8294	0.6406	0.4172	0.8220

SVC

Dataset	Accuracy
Training Set	80.38 %
Validation Set	64.71%
Test Set	76.47 %

➤ Model Overview

- Algorithm: Support Vector Classifier (SVC)
- Objective: Classify using optimal decision boundaries
- Tuned with: GridSearchCV (3-fold cross-validation)

➤ Hyperparameter Tuning

- Parameters tested:
 - C: [0.1, 1, 10]
 - Kernel: ['rbf', 'linear']
- Best Parameters: C = 1, kernel = 'rbf'
- Best CV Score: 71.19%

➤ Model Performance

- Training Set: 80.38%
- Validation Set: 64.71%
- Test Set: 76.47%

➤ Key Insights

- RBF kernel outperformed linear kernel
- Good generalization to the test set
- Slight overfitting, but within reasonable limits
- Room for improvement via:
 - Feature scaling

Random Forest

Baseline Model

Initial Parameters:

- n_estimators=500
- max_depth=12

Results:

- Training accuracy: 0.9241
- Validation accuracy: 0.5882

Optimized Model

Hyperparameter Tuning: (GridSearchCV)

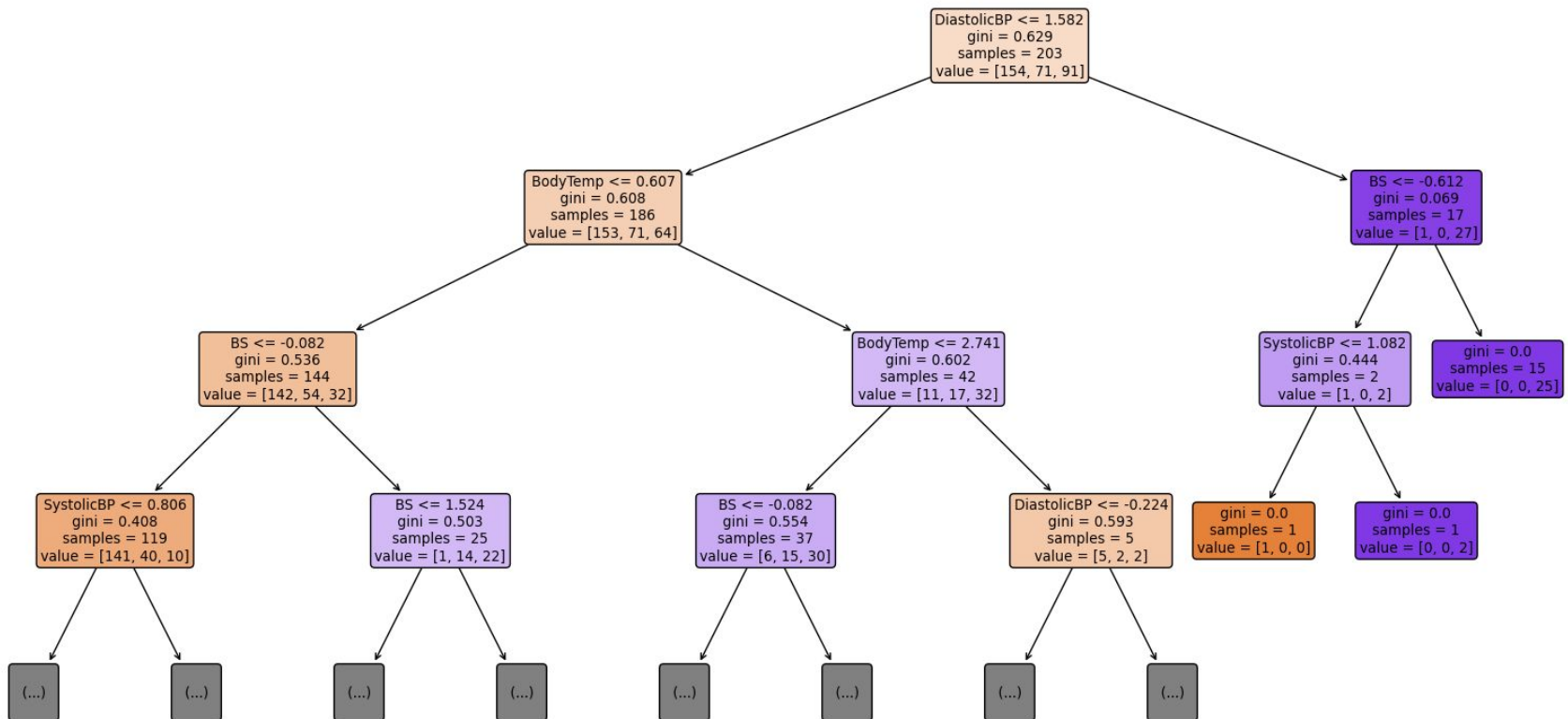
- Best parameters Found:
 - criterion = entropy
 - max_depth = 7
 - min_samples_leaf = 2
 - min_samples_split = 2

Results:

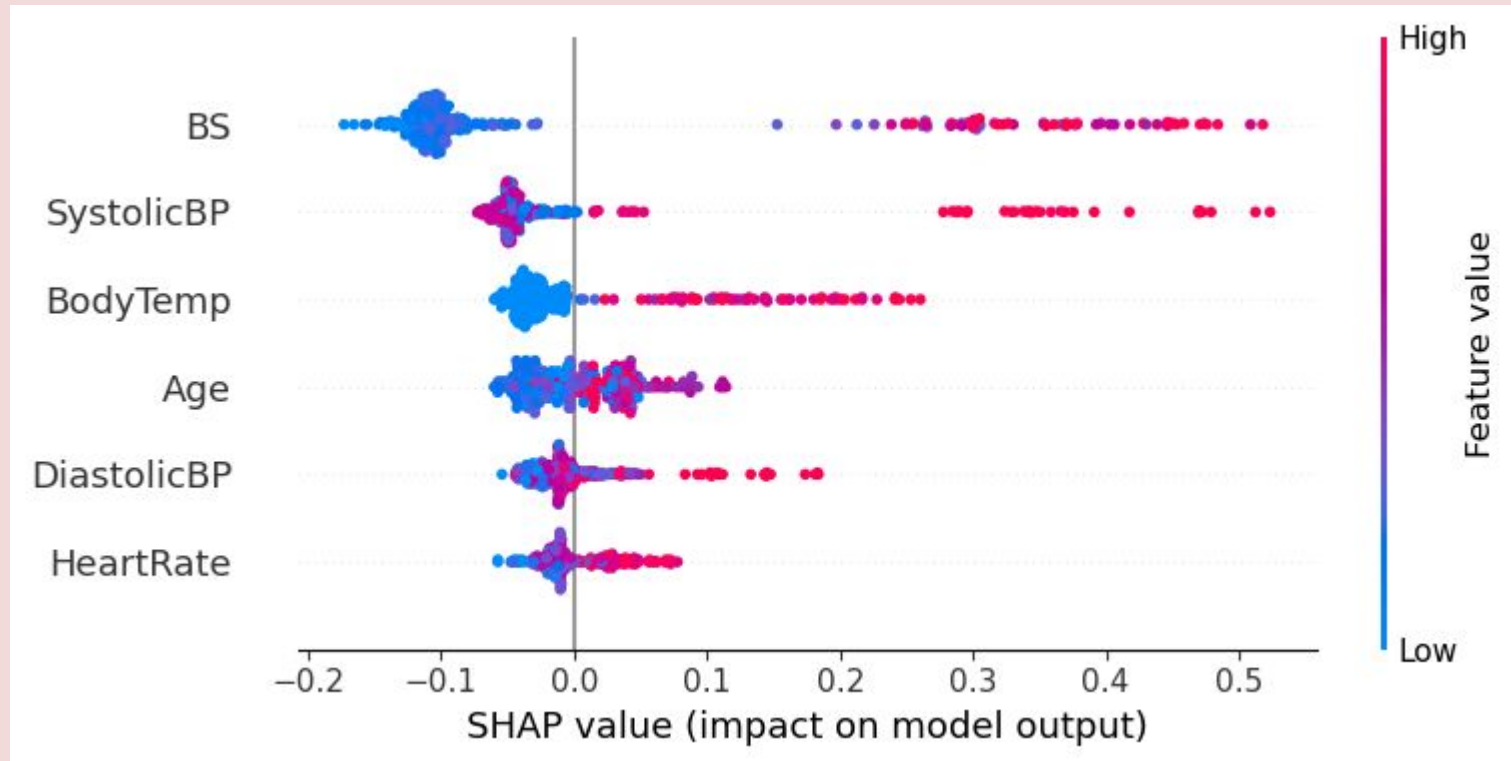
- Training accuracy: 0.8038
- Validation accuracy: 0.6912
- Test accuracy: 0.7647

Random Forest Experimentation

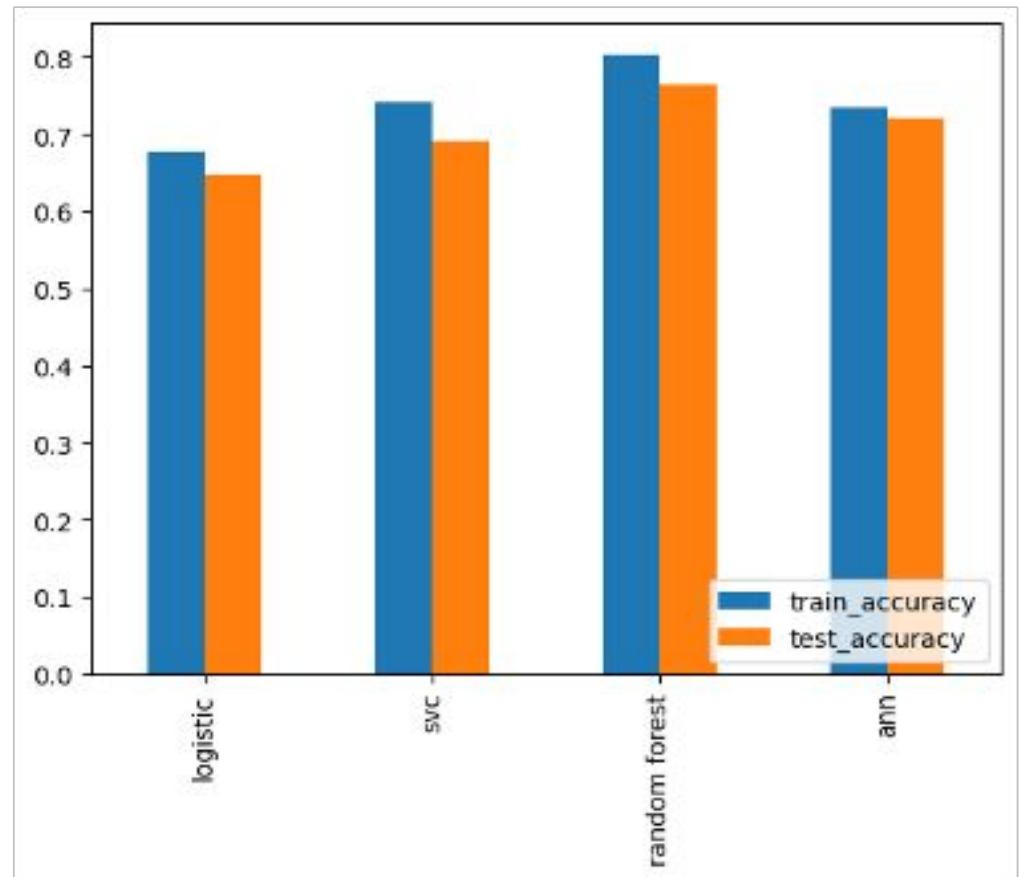
Visualization of Last Tree from the Random Forest



Feature Importance Interpretation (Random Forest)



Summary



Limitations and Considerations

- Possible limitations:
 - Data is from rural area and may not be very generalizable
 - Dataset was small (~1000 samples) so models tend to overfit with complexity
 - We were intentional in our feature engineering to not create features that did not fit with the medical literature.

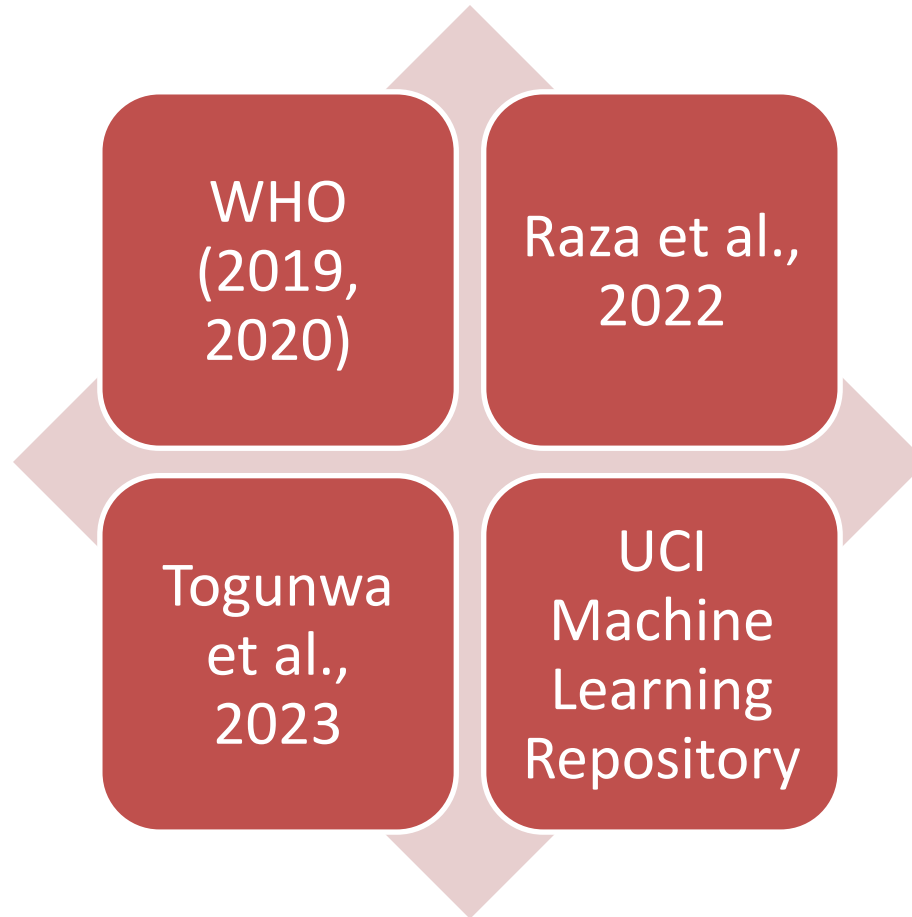


Recommendations



1. Increase Dataset Size
 - Collaborate with additional hospitals and clinics across diverse geographic regions to boost sample volume and reduce population bias.
 - Explore data-sharing agreements or open-access clinical datasets to expand the training base.
2. Expand Population Diversity
 - Ensure data is collected across different age groups, ethnicities, socioeconomic statuses, and urban vs rural populations to improve model generalizability.

References



Code and Contributions

Code

- GitHub Repo:
https://github.com/gowrath/maternal_health_risk

Contributions:

	Eric	John	Nazia
Theoretical/Market Research	X	X	X
EDA	X	X	X
Data Cleaning	X	X	X
Feature Engineering	X	X	X
Model Exploration	Logistic Regression, ANN	Random Forest	SVC
Presentation Slides	X	X	X